Single-point accuracy across three PDEs

RMSE above RelMAE · 32 frozen query states · bold = best valid value

Method	Full-Rank Log-Sum-Exp HJB					Fourier–Tanh Coupled PDE					Joint Nonseparable PDE				
	20	50	100	200	400	20	50	100	200	400	20	50	100	200	400
Tangent-Actor	**7.13e-4**	**1.42e-3**	**1.59e-3**	**4.17e-3**	**3.63e-3**	**4.25e-3**	**4.32e-3**	**3.72e-3**	**8.69e-3**	**1.40e-2**	**1.18e-3**	**1.06e-3**	**1.04e-3**	**1.22e-3**	**1.24e-3**
	7.84e-4	**8.76e-4**	**9.04e-4**	**1.33e-3**	**1.25e-3**	**6.06e-3**	**7.25e-3**	**2.33e-3**	**1.98e-2**	**3.15e-2**	**1.68e-3**	**1.54e-3**	**1.49e-3**	**1.70e-3**	**1.77e-3**
FH-MLP	8.56e-2	2.71e-1	5.89e-1	1.93e0	3.19e0	3.60e-2	4.23e-2	1.33e-1	1.43e-1	2.93e-1	1.14e-2	2.56e-2	5.15e-2	1.02e-1	2.13e-1
	1.16e-1	2.03e-1	4.32e-1	7.49e-1	1.24e0	4.98e-2	7.21e-2	1.07e-1	3.56e-1	6.37e-1	1.46e-2	3.44e-2	7.39e-2	1.41e-1	3.00e-1
Deep BSDE	1.13e-1	2.53e-1	2.82e-1	3.72e-1	3.24e-1	4.38e-2	7.79e-2	1.13e-1	7.52e-2	7.91e-2	2.59e-3	3.76e-3	3.48e-2	*NF*	*NF*
	1.58e-1	1.94e-1	2.11e-1	1.49e-1	1.30e-1	6.95e-2	1.45e-1	9.37e-2	1.94e-1	1.73e-1	3.35e-3	4.46e-3	1.77e-2	*NF*	*NF*

NF: numerical failure in the frozen Deep BSDE configuration.